WINMO

TECHNICAL WHITEPAPER V1.1

Leveraged ETFs for Every Asset Onchain.

SPOT. LEVERAGED. INVERSE.

Non-liquidatable tokens across stocks, indices, commodities, forex, and crypto.
2x, 3x long. -1x, -2x inverse. No KYC. No liquidation.
The leveraged layer for all of DeFi.

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Winmo Foundation · March 2026

TABLE OF CONTENTS

1 **Spot. Leveraged. Inverse. Every Asset Onchain.** 4
- 1.1 The leveraged layer for all of DeFi. 4

2 **1. Abstract** 4

3 **2. Vision** 4

4 **3. Market Opportunity & Business Case** 5
- 4.1 3.1 A $60B Market with Zero Onchain Equivalent 5
- 4.2 3.2 Why Lending-Based Leverage Failed 5
- 4.3 3.3 Why Now -The Hyperliquid Catalyst 6
- 4.4 3.4 Revenue Model & Unit Economics 6
- 4.5 3.5 Competitive Landscape 7

5 **4. Technical Introduction** 8
- 5.1 4.1 How It Works 8
- 5.2 4.2 Four-Layer Defense System 8

6 **5. Mechanism Design -Long (2x)** 8
- 6.1 5.1 Core Concept 8
- 6.2 5.2 Position Mechanics 8
- 6.3 5.3 Why Liquidation Cannot Occur (Long 2x) 9
- 6.4 5.4 What Users Hold 10

7 **6. Mechanism Design -Inverse (-1x)** 11
- 7.1 6.1 Core Concept 11
- 7.2 6.2 Inverse Position Mechanics 12
- 7.3 6.3 Why Inverse Products Cannot Be Liquidated 12
- 7.4 6.4 Inverse-Specific Considerations 13
- 7.5 6.5 Inverse Vault Smart Contract Differences 13

8 **7. RWA Asset Coverage** 14
- 8.1 7.1 RWA Asset Characteristics 14
- 8.2 7.2 RWA Safety Advantage 14
- 8.3 7.3 RWA Total Annual Drag 14
- 8.4 7.4 Off-Market Hours Handling 15

9 **8. Crypto Asset Coverage** 15
- 9.1 8.1 Crypto Asset Characteristics 15
- 9.2 8.2 Safety Parameters 16
- 9.3 8.3 Crypto Crash Simulation: BTC -37.5% in 24 Hours (March 2020) 16
- 9.4 8.4 Crypto Inverse Products: Extra Considerations 17
- 9.5 8.5 Crypto Execution on Hyperliquid 17

10 **9. Rebalancing Engine** 18
- 10.1 9.1 Layer 1: Scheduled Daily Rebalance 18
- 10.2 9.2 Layer 2: Threshold-Based Intraday Rebalance 19
- 10.3 9.3 Layer 3: Emergency Rebalance (Ripcord) 19
- 10.4 9.4 Layer 4: Circuit Breaker (Nuclear Option) 20
- 10.5 9.5 Defense-in-Depth Summary 20
- 10.6 9.6 ADL (Auto-Deleveraging) Handling 20

10.7 9.7 Continuous Market Coverage . . . 21

11 10. Execution Layer: Hyperliquid **21**
11.1 10.1 Hyperliquid Overview . . . 21
11.2 10.2 Integration Approaches . . . 21
11.3 10.3 Fee Impact . . . 22
11.4 10.4 Execution Routing . . . 22

12 11. Smart Contract Architecture **22**
12.1 11.1 Contract Overview . . . 22
12.2 11.2 LeveragedVault.sol -Core Contract . . . 23
12.3 11.3 Keeper Infrastructure . . . 26

13 12. Mathematical Framework **27**
13.1 12.1 Leveraged Return Formula . . . 27
13.2 12.2 Volatility Decay (Variance Drain) . . . 27
13.3 12.3 Funding Rate Impact . . . 28
13.4 12.4 Total Expected Annual Drag . . . 28

14 13. Protocol Risk Analysis **29**
14.1 13.1 Smart Contract Risk . . . 30
14.2 13.2 Price / Execution Risk . . . 30
14.3 13.3 Execution / Rebalancing Risk . . . 31
14.4 13.4 Counterparty / Protocol Risk . . . 31
14.5 13.5 Economic / Market Risk . . . 32
14.6 13.5a Inverse-Specific Risk . . . 33
14.7 13.5b Crypto-Specific Risk . . . 34
14.8 13.6 Bridge / Cross-Chain Risk . . . 34
14.9 13.7 Governance / Operational Risk . . . 35
14.1013.8 Risk Matrix Summary . . . 35

15 14. Fee Structure & Economics **35**
15.1 14.1 User-Facing Fees . . . 35
15.2 14.2 Internal Costs (Absorbed by NAV) . . . 35
15.3 14.3 Revenue Model at Scale . . . 36

16 15. Governance & Operations **36**
16.1 15.1 Governance-Controlled Parameters . . . 36
16.2 15.2 Operational Roles . . . 36

17 16. Roadmap **37**

18 17. Conclusion **37**

19 18. References **38**

20 19. Disclaimer **38**

Spot. Leveraged. Inverse. Every Asset Onchain.

The leveraged layer for all of DeFi.

March 2026

1. Abstract

This paper presents the Winmo Leveraged ETF Protocol -a permissionless, non-custodial system that creates ERC-20 tokens providing daily **2x, 3x long and -1x, -2x inverse** exposure to real-world asset prices (stocks, commodities, indices, forex) and cryptocurrency (BTC, ETH, SOL, and more) using perpetual futures positions on Hyperliquid as the underlying execution layer.

The critical design property is that the product **cannot be liquidated** at the user level. While the underlying mechanism uses leveraged perpetual futures on third-party DEXs, the vault's daily rebalancing algorithm ensures that the effective margin ratio never approaches the liquidation threshold of the underlying protocol. A multi-layered safety system -comprising scheduled rebalancing, emergency rebalancing, ripcord deleveraging, and circuit breakers -provides defense-in-depth against all market scenarios including black swan events.

The protocol offers four leverage tiers: **Long 2x** (e.g., sAAPL-2x), **Long 3x** (e.g., sAAPL-3x), **Inverse -1x** (e.g., sAAPL-1xS), and **Inverse -2x** (e.g., sAAPL-2xS). Long tokens profit when the underlying rises; inverse tokens profit when it falls. All tiers are non-liquidatable and use the same four-layer defense system, with mirrored mechanics -long vaults hold long perp positions while inverse vaults hold short perp positions.

The trade-off for non-liquidatability is **volatility decay** -a mathematically inherent property of all daily-rebalanced leveraged products, including traditional leveraged ETFs like ProShares TQQQ and Direxion SPXL. For inverse products, an additional consideration is that the **underlying asset has theoretically unlimited upside**, making long-term holding of inverse products particularly costly in trending bull markets.

2. Vision

The Leveraged ETF Protocol for Every Asset Onchain.

Permissionless access to every asset onchain. Non-liquidatable tokens across stocks, indices, commodities, forex, and crypto. Spot. 2x, 3x long. -1x, -2x inverse. No KYC. No liquidation.

Winmo exists because onchain traders deserve the same products that TradFi traders have had for decades -but better. Leveraged ETFs are among the most traded products in traditional finance. ProShares TQQQ, Direxion SPXL. Billions in daily volume. Yet in DeFi, this product category barely exists. There is zero on-chain access to leveraged S&P 500, Apple, Gold, or EUR/USD. No product provides permissionless, 24/7, non-liquidatable leveraged exposure to real-world asset prices or crypto.

We talked to the people who actually trade onchain every day -crypto-native traders who move size, take risk, and live on perpetual futures. What we heard was consistent across dozens of conversa-

tions:

They like stocks but want greater returns. SPY returns 8-10% a year. For a trader used to 50-100% swings in crypto, that is not interesting. But 2x leveraged SPY? That changes the math entirely. The same traders who would never buy SPY would absolutely buy sSPX-2x.

They want non-KYC access. These are onchain traders. They have wallets, they have USDC, they are ready to deploy. But every platform that offers stocks or RWA requires KYC, separate accounts, and days of onboarding. They want to deposit USDC and get exposure. Permissionless. No signup. No identity verification. Just a wallet and a token.

They do not want to get liquidated. Every onchain trader has a liquidation story. A position wiped out by a wick, a cascade, a funding spike. They want leveraged exposure without the risk of losing everything in a flash crash. They want to hold a token that gives them 2x exposure to Apple or Bitcoin without worrying about margin calls at 3am.

Winmo fixes all three. Leveraged returns on every asset class. Fully permissionless, no KYC. And non-liquidatable by design with a four-layer safety system. Deposit USDC, hold an ERC-20 token like sSPX-2x or sAAPL-2x, and get the leveraged exposure that TradFi traders have had for years -except onchain, 24/7, and without the liquidation risk.

3. Market Opportunity & Business Case

3.1 A $60B Market with Zero Onchain Equivalent

Traditional leveraged ETFs are among the most traded products in TradFi. The combined AUM of leveraged and inverse ETFs exceeds **$60 billion**. ProShares TQQQ alone manages over $20B. These products provide 2-3x daily leveraged exposure to indices and are used by millions of traders every day.

In DeFi, this product category barely exists. IndexCoop created ETH2x-FLI for Ethereum exposure using Aave/Compound lending. It peaked at **$200M AUM** and proved massive demand. But it was ultimately sunset due to structural costs -limited to 2 assets, could only go long, and was dependent on Aave governance parameters it couldn't control.

The demand is validated. The architecture was wrong. Winmo fixes the architecture.

3.2 Why Lending-Based Leverage Failed

IndexCoop's FLI used Aave lending: deposit ETH, borrow USDC, buy more ETH, redeposit, repeat until ~2x exposure. This approach has inherent limitations that cannot be fixed with better engineering:

- Gas costs consumed 1-3% of NAV annually (3-5 L1 transactions per rebalance).
- Aave governance controlled LTV ratios, interest rates, and could freeze markets. FLI had no control over its own leverage engine.
- The May 2021 crash nearly liquidated the product. $10M+ rebalances moved markets against FLI on Uniswap.
- Borrow rates spiked to 90%+ in DeFi summer, creating extreme and unpredictable leverage costs.
- FLI's operational costs were 5-10x higher than Winmo's for the same product.

The failure was architectural, not operational. Lending-based leverage is structurally limited to a handful of crypto assets, can only go long, and cannot support real-world assets.

3.3 Why Now -The Hyperliquid Catalyst

Hyperliquid now supports both crypto and RWA perpetual futures (stocks, indices, commodities, forex) with deep liquidity, sub-second price updates, and low fees. Hyperliquid's HIP-3 upgrade enables permissionless perpetual futures for any asset -22 equity pairs, 12 commodity and forex pairs, and 8+ major crypto pairs are already live with deep liquidity. All RWA pairs trade 24/7.

Perps solve every structural problem that killed lending-based leverage:

Dimension	Lending-Based (FLI)	Perp-Based (Winmo)
Capital efficiency	Low -overcollateralized	High -direct leverage
Asset coverage	2 assets (ETH, BTC)	Any asset with a perp market
Rebalancing cost	Multi-step lending loop	Single position adjustment
Gas cost	High (L1 Ethereum)	Near zero (Arbitrum / HL L1)
Swap cost	0.3% per rebalance (Uniswap)	0.03% per rebalance (Hyperliquid)
Governance dependency	Aave controls parameters	Hyperliquid execution only
Direction	Long only	Long AND inverse
Asset classes	Crypto only	Crypto + RWA (stocks, commodities, indices, forex)

For the first time, it is possible to build leveraged tokens for Apple, the S&P 500, Gold, EUR/USD, Bitcoin, and dozens more -all from one execution layer, all permissionless, all non-liquidatable.

3.4 Revenue Model & Unit Economics

Revenue comes from every layer of the protocol:

Revenue Source	Rate	Description
Management fee	1.0% annual	Accrued daily to NAV across all vaults
Mint/Redeem fees	0.1% each	Collected on every deposit and withdrawal
Maker rebates	Variable	Protocol share of Hyperliquid maker rebates

Projected revenue at scale:

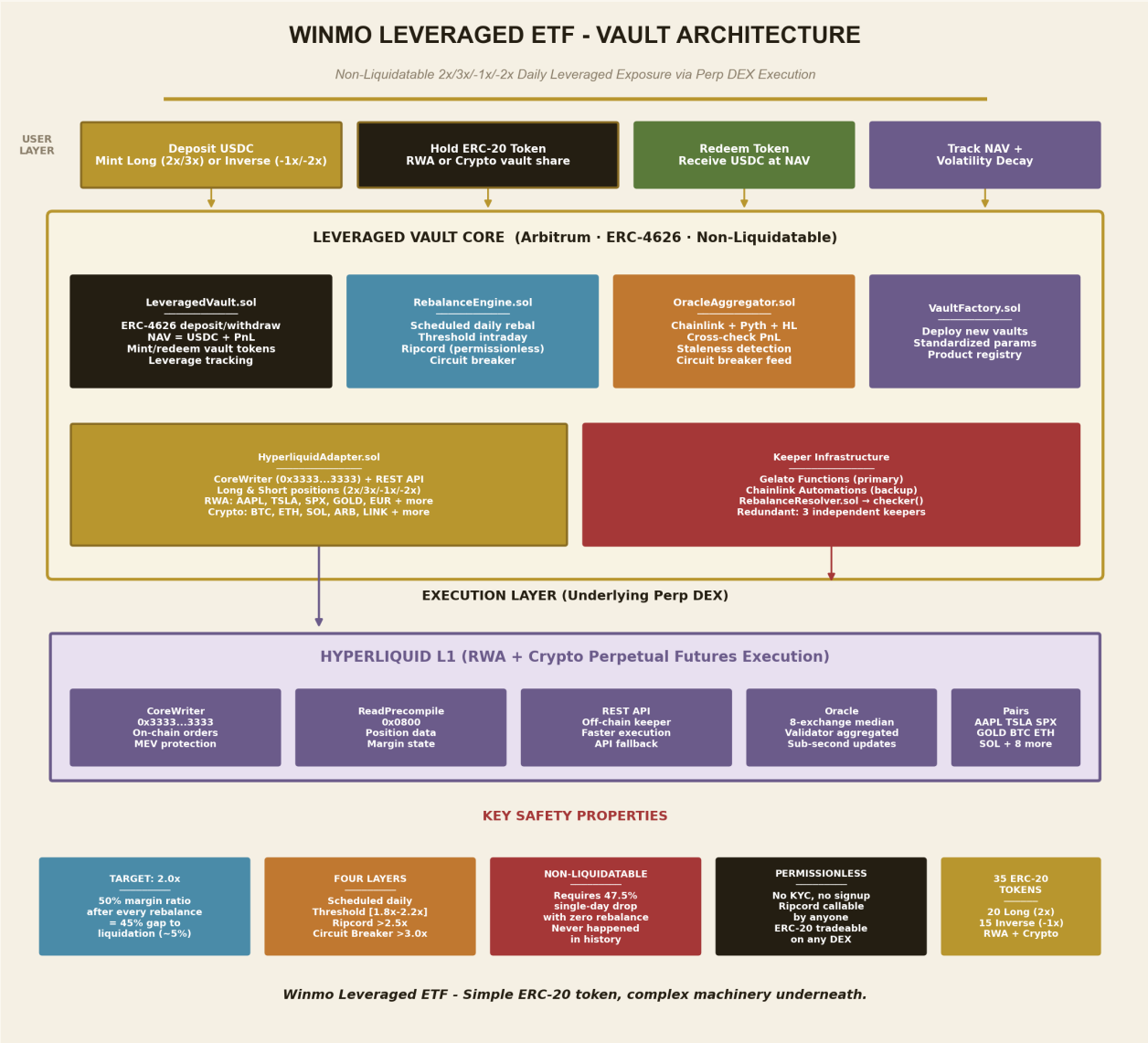
AUM	Management Fee	Mint/Redeem Fees (10% turnover/mo)	Total Revenue
$10M	$100K	$240K	**$340K**
$50M	$500K	$1.2M	**$1.7M**
$100M	$1M	$2.4M	**$3.4M**
$500M	$5M	$12M	**$17M**

Unit economics improve with scale -each new vault reuses the same infrastructure (keeper network, Hyperliquid integration, smart contract suite). The marginal cost of adding a new leveraged token is near zero once the vault factory is deployed.

3.5 Competitive Landscape

Protocol	Assets	Leverage Tiers	Permissionles	Non-Liquidatable	Status
ProShares TQQQ/SQQQ	QQQ only	3x/-3x	No (KYC)	N/A (TradFi)	Active
Direxion SPXL/SPXS	S&P 500 only	3x/-3x	No (KYC)	N/A (TradFi)	Active
IndexCoop ETH2x-FLI	ETH only	2x	Yes	No	**Sunset**
Winmo	**Stocks, indices, commodi-ties, forex, crypto**	**2x, 3x, -1x, -2x**	**Yes**	**Yes**	**Building**

No product in DeFi or TradFi offers non-liquidatable leveraged exposure across stocks, indices, commodities, forex, and crypto -long and inverse, permissionless, 24/7. Winmo is building the leveraged layer for all of DeFi.



4. Technical Introduction

4.1 How It Works

Deposit USDC. Hold a token. Redeem at NAV.

1. **Deposit:** User deposits USDC into a vault and receives an ERC-20 token (e.g., sAAPL-2x).
2. **Execution:** The vault opens a leveraged perpetual futures position on Hyperliquid.
3. **Rebalancing:** Daily rebalancing resets leverage to the target (2x, 3x, -1x, or -2x). Four safety layers prevent liquidation.
4. **Redemption:** User redeems the token at any time for USDC at current NAV.

For inverse tokens, the vault opens a SHORT perp position instead of long. At -1x, the vault operates at 1.0x absolute leverage with 100% margin ratio, making it inherently safer than 2x long products.

The user experience is a simple ERC-20 token. Complex machinery underneath. No perp account, no margin management, no liquidation risk. Hold the token in any wallet. Trade it on any DEX. The vault handles everything.

4.2 Four-Layer Defense System

Non-liquidatable by design. Four layers of protection:

Layer	Trigger	Action	
Layer 1: Scheduled Daily Rebalance	Every day at market close	Resets leverage to target. Margin ratio resets to 50%.	
Layer 2: Threshold Intraday	Leverage deviates beyond [1.8x, 2.2x] band	Keepers trigger intraday rebalance. Catches 5-10% intraday moves.	
Layer 3: Emergency Ripcord	Underlying drops >12.5% since last rebalance, leverage >2.5x	Permissionless -anyone can call. Deleverages to 1.5x. Caller receives USDC reward.	
Layer 4: Circuit Breaker	Leverage exceeds 3.0x (underlying dropped >25%)	Vault closes all positions. Holds only USDC. Governance vote to resume.	

After every rebalance, margin resets to 50%. Liquidation requires margin to fall below 5% -meaning a 47.5% single-day drop with zero rebalancing. No major asset in recorded financial history has dropped 47.5% in a single day.

5. Mechanism Design -Long (2x)

5.1 Core Concept

A Winmo Leveraged ETF vault (long 2x):

1. Accepts USDC deposits from users.
2. Mints an ERC-20 token (e.g., sAAPL-2x) representing a share of the vault's NAV.
3. Opens a 2x leveraged perpetual futures position on Hyperliquid.
4. Rebalances daily (and on emergency triggers) to maintain exactly 2x leverage.
5. Allows permissionless minting and redemption at NAV.

5.2 Position Mechanics

Initial state:

```
Vault deposits:     $10,000,000 USDC
Target leverage:    2.0x
Perp position:      LONG AAPL-USD, $20,000,000 notional
Collateral posted:  $10,000,000 USDC to Hyperliquid
Effective leverage:  2.0x
Margin ratio:       50% (collateral / notional)
```

After AAPL rises 5% (before rebalance):

```
Unrealized PnL:     +$1,000,000 (5% x $20M notional)
Vault NAV:          $11,000,000
Position notional:  $21,000,000 (price moved up)
Current leverage:   $21,000,000 / $11,000,000 = 1.909x
                    (leverage has DECREASED because gains increased NAV)
Action needed:      BUY more to restore 2.0x
New target notional: $22,000,000 (2.0 x $11M)
Delta to execute:    +$1,000,000 notional (increase position)
```

After AAPL drops 5% (before rebalance):

```
Unrealized PnL:     -$1,000,000
Vault NAV:          $9,000,000
Position notional:  $19,000,000
Current leverage:   $19,000,000 / $9,000,000 = 2.111x
                    (leverage has INCREASED because losses decreased NAV)
Action needed:      SELL to restore 2.0x
New target notional: $18,000,000 (2.0 x $9M)
Delta to execute:    -$1,000,000 notional (decrease position)
```

This is the key safety property: After losses, the vault SELLS (reduces exposure). This prevents leverage from spiraling upward and approaching the liquidation threshold.

5.3 Why Liquidation Cannot Occur (Long 2x)

Liquidation on Hyperliquid occurs when margin ratio drops below maintenance margin (approximately half of initial margin at max leverage). For a 2x position, maintenance margin is approximately 2.5% of notional, meaning margin ratio must drop below 2.5%.

With daily rebalancing, the vault's margin ratio resets to ~50% every day. The distance from 50% to 2.5% (liquidation) is a 95% loss of NAV -which requires approximately a **47.5% single-day drop in the underlying asset** without any intraday rebalance.

Historical single-day drops for context:

Asset	Largest Single-Day Drop	2x Loss	Would Trigger Liquidation?
S&P 500	-20.47% (Oct 1987)	-40.94%	**No** (margin still 9%)
Apple	-12.35% (Sep 2020)	-24.70%	**No** (margin still 25.3%)
Tesla	-21.06% (Sep 2020)	-42.12%	**No** (margin still 7.9%)
Gold	-5.72% (Nov 2020)	-11.44%	**No** (margin still 38.6%)
EUR/USD	-3.47% (Mar 2015)	-6.94%	**No** (margin still 43%)
Bitcoin	-37.5% (Mar 2020)	-75.0%	Intraday rebalancing prevents (see Section 7)
Ethereum	-43.0% (Mar 2020)	-86.0%	Intraday rebalancing prevents (see Section 7)

No major real-world asset has ever dropped 47.5% in a single day. The worst single-day drop in the S&P 500's entire history (-20.47%, Black Monday 1987) would still leave the vault with

9% margin -well above the 2.5% liquidation threshold. For RWA products, the four-layer defense system provides an extraordinary margin of safety.

For crypto assets: Crypto has experienced single-day drops exceeding 30%, making intraday rebalancing layers critical for crypto vaults. See Section 7 for crypto-specific safety parameters and crash simulations.

5.4 What Users Hold

Each Winmo ETF product is an **ERC-20 token** deployed on Arbitrum. Every Long (2x) product has a corresponding Inverse (-1x) product:

Long 2x Products (profit when underlying rises):

Token	Underlying	Direction	Execution	Class
sAAPL-2x	Apple	2x Long	Hyperliquid	Equities
sTSLA-2x	Tesla	2x Long	Hyperliquid	Equities
sNVDA-2x	Nvidia	2x Long	Hyperliquid	Equities
sAMZN-2x	Amazon	2x Long	Hyperliquid	Equities
sMSFT-2x	Microsoft	2x Long	Hyperliquid	Equities
sMETA-2x	Meta	2x Long	Hyperliquid	Equities
sSPX-2x	S&P 500	2x Long	Hyperliquid	Indices
sNDX-2x	Nasdaq 100	2x Long	Hyperliquid	Indices
sGOLD-2x	Gold	2x Long	Hyperliquid	Commodities
sOIL-2x	WTI Crude	2x Long	Hyperliquid	Commodities
sSLVR-2x	Silver	2x Long	Hyperliquid	Commodities
sEUR-2x	EUR/USD	2x Long	Hyperliquid	Forex
sBTC-2x	Bitcoin	2x Long	Hyperliquid	Crypto
sETH-2x	Ethereum	2x Long	Hyperliquid	Crypto
sSOL-2x	Solana	2x Long	Hyperliquid	Crypto
sARB-2x	Arbitrum	2x Long	Hyperliquid	Crypto
sLINK-2x	Chainlink	2x Long	Hyperliquid	Crypto
sAVAX-2x	Avalanche	2x Long	Hyperliquid	Crypto
sDOGE-2x	Dogecoin	2x Long	Hyperliquid	Crypto
sSUI-2x	Sui	2x Long	Hyperliquid	Crypto

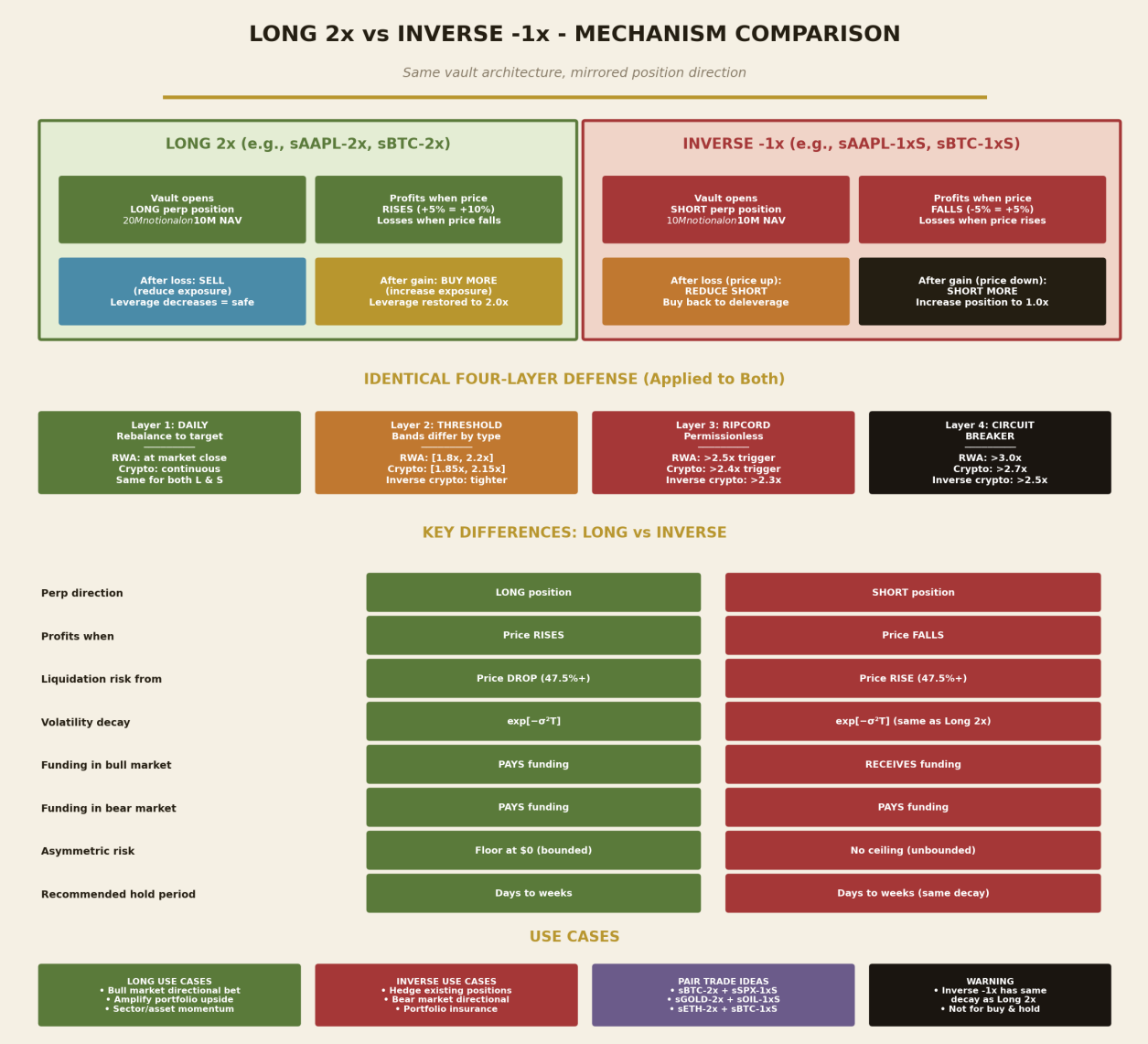
Inverse -1x Products (profit when underlying falls):

Token	Underlying	Direction	Execution	Class
sAAPL-1xS	Apple	-1x Inverse	Hyperliquid	Equities
sTSLA-1xS	Tesla	-1x Inverse	Hyperliquid	Equities
sNVDA-1xS	Nvidia	-1x Inverse	Hyperliquid	Equities
sSPX-1xS	S&P 500	-1x Inverse	Hyperliquid	Indices
sNDX-1xS	Nasdaq 100	-1x Inverse	Hyperliquid	Indices
sGOLD-1xS	Gold	-1x Inverse	Hyperliquid	Commodities
sEUR-1xS	EUR/USD	-1x Inverse	Hyperliquid	Forex
sBTC-1xS	Bitcoin	-1x Inverse	Hyperliquid	Crypto
sETH-1xS	Ethereum	-1x Inverse	Hyperliquid	Crypto
sSOL-1xS	Solana	-1x Inverse	Hyperliquid	Crypto
sARB-1xS	Arbitrum	-1x Inverse	Hyperliquid	Crypto
sLINK-1xS	Chainlink	-1x Inverse	Hyperliquid	Crypto
sAVAX-1xS	Avalanche	-1x Inverse	Hyperliquid	Crypto
sDOGE-1xS	Dogecoin	-1x Inverse	Hyperliquid	Crypto
sSUI-1xS	Sui	-1x Inverse	Hyperliquid	Crypto

The protocol offers 20 Long and 15 Inverse products across 5 asset classes (equities, indices, com-

modities, forex, crypto). All execute on Hyperliquid.

Token price tracks vault NAV per share. Users can mint (deposit USDC) and redeem (burn tokens, receive USDC) at any time.



6. Mechanism Design -Inverse (-1x)

6.1 Core Concept

An Inverse vault is the mirror image of a Long vault. Instead of opening a LONG perp position, the vault opens a **SHORT** perp position. The vault profits when the underlying price **falls** and loses when it **rises**.

A Winmo Inverse ETF vault:

1. Accepts USDC deposits from users.
2. Mints an ERC-20 token (e.g., sAAPL-1xS) representing a share of the vault's NAV.
3. Opens a **2x leveraged SHORT** perpetual futures position on Hyperliquid.
4. Rebalances daily (and on emergency triggers) to maintain exactly -1x leverage.

5. Allows permissionless minting and redemption at NAV.

6.2 Inverse Position Mechanics

Initial state:

```
Vault deposits:     $10,000,000 USDC
Target leverage:    -1.0x (SHORT)
Perp position:      SHORT SPX-USD, $10,000,000 notional
Collateral posted:  $10,000,000 USDC to Hyperliquid
Effective leverage:  -1.0x
Margin ratio:       100% (collateral / notional)
```

After S&P 500 drops 5% (vault profits):

```
Unrealized PnL:     +$500,000 (SHORT profits when price falls)
Vault NAV:          $10,500,000
Short notional:     $9,500,000 (price moved down, short notional decreased)
Current leverage:   $9,500,000 / $10,500,000 = -0.905x
                    (leverage magnitude DECREASED - NAV grew, notional shrank)
Action needed:      SHORT MORE to restore -1.0x
New target notional: $10,500,000 (1.0 x $10.5M)
Delta to execute:    +$1,000,000 notional (increase short position)
```

After S&P 500 rises 5% (vault loses):

```
Unrealized PnL:     -$500,000 (SHORT loses when price rises)
Vault NAV:          $9,500,000
Short notional:     $10,500,000 (price moved up, short notional increased)
Current leverage:   $10,500,000 / $9,500,000 = -1.105x
                    (leverage magnitude INCREASED - NAV shrank, notional grew)
Action needed:      REDUCE SHORT to restore -1.0x
New target notional: $9,500,000 (1.0 x $9.5M)
Delta to execute:    -$1,000,000 notional (decrease short position / partial close)
```

Key safety property (identical to long): After losses (price rises for inverse), the vault REDUCES its short (buys back). This prevents leverage from spiraling toward liquidation.

6.3 Why Inverse Products Cannot Be Liquidated

The same four-layer defense system protects inverse vaults:

- **Daily rebalance** resets margin ratio to 50%.
- **Threshold rebalance** triggers at 2.2x magnitude (if underlying rises ~5%).
- **Ripcord** at 2.5x magnitude (if underlying rises ~12.5%).
- **Circuit breaker** at 3.0x magnitude (if underlying rises ~25%).

For the inverse vault, liquidation risk comes from the underlying **rising** sharply. A SHORT position on Hyperliquid is liquidated when margin drops below ~2.5%. With 50% margin, this requires a **47.5% single-day RISE** in the underlying.

Historical single-day rises for context:

Asset	Largest Single-Day Rise	-1x Loss	Would Trigger Liquidation?
S&P 500 (reference)	+11.58% (Oct 2008)	-11.58%	No (margin still 38.4%)
Tesla (reference)	+19.64% (Jun 2020)	-19.64%	No (margin still 30.4%)
Bitcoin	+42.0% (Feb 2021)	-42.0%	No (margin still 8.0%)
Ethereum	+30.0% (Jul 2021)	-30.0%	No (margin still 20.0%)

Note: Traditional assets like the S&P 500 and Tesla are shown for comparison -no major tradi-

tional asset has risen 47.5% in a single day.

For crypto assets: Crypto can rally 30-40%+ in a day. Inverse crypto products (sBTC-1xS, sETH-1xS) are higher risk and rely heavily on intraday rebalancing layers. The protocol enforces **tighter bands** for inverse crypto vaults: [1.85x, 2.15x] threshold with 3-minute keeper checks (see Section 5).

6.4 Inverse-Specific Considerations

Funding rate advantage: In bull markets, short positions on perp DEXs typically **receive** funding (longs pay shorts). This means inverse products can have **lower** total cost than long products during bull markets. Conversely, in bear markets, shorts pay longs.

Asymmetric risk: Underlying assets have a floor of $0 but **no ceiling**. A long 2x product's maximum loss is 100% of NAV (underlying goes to $0). An inverse -1x product's maximum loss is also 100% of NAV (underlying doubles -100% rise), but the probability of a 100% rise over a longer period is much higher than a drop to zero. **Inverse products have higher long-term risk in trending bull markets.**

Volatility decay: Inverse -1x products experience the **same** volatility decay as long 2x products. The decay formula is identical: `Decay = exp[--`σ^2` × T]`. However, the combination of volatility decay AND trending markets against the position makes holding inverse products for extended periods costly.

Use cases for inverse products:

- **Hedging:** Hold sAAPL-1xS to hedge an existing Apple stock position.
- **Bear market trading:** Profit from market downturns without needing a perp account.
- **Pair trading:** Long sBTC-2x + Short sSPX-1xS for BTC vs SPX relative value.
- **Volatility harvesting:** Mint/redeem around rebalance windows.

6.5 Inverse Vault Smart Contract Differences

The inverse vault uses the same `LeveragedVault.sol` contract with `isLong = false`:

```
// In LeveragedVault.sol
bool public isLong;  // true = 2x Long, false = -1x Inverse

function _rebalanceToTarget(uint256 targetLev) internal {
    uint256 nav = totalAssets();
    uint256 currentNotional = dexAdapter.getPositionNotional(address(this));
    uint256 targetNotional = (nav * targetLev) / 1e18;

    if (targetNotional > currentNotional) {
        uint256 delta = targetNotional - currentNotional;
        if (_isAboveMinThreshold(delta, currentNotional)) {
            if (isLong) {
                dexAdapter.increasePosition(delta);    // Buy more (long)
            } else {
                dexAdapter.increaseShort(delta);       // Short more (inverse)
            }
        }
    } else {
        uint256 delta = currentNotional - targetNotional;
        if (_isAboveMinThreshold(delta, currentNotional)) {
            if (isLong) {
                dexAdapter.decreasePosition(delta);    // Sell some (long)
            } else {
                dexAdapter.decreaseShort(delta);       // Buy back (inverse)
            }
        }
    }
}
```

The rebalancing direction logic, ripcord, and circuit breaker remain identical -only the perp direction changes.

7. RWA Asset Coverage

Real-world assets -equities, indices, commodities, and forex -are the primary focus of the Winmo protocol. These assets represent over $250 trillion in global value and are the core market opportunity.

7.1 RWA Asset Characteristics

Property	Equities	Indices	Commodities	Forex
Trading hours	Market hours + extended	Market hours	Market hours	24/5
Annualized volatility	20-55%	12-20%	15-30%	5-12%
Largest single-day move	-21% (Tesla)	-20.47% (SPX)	-9% (Oil)	-3.5% (EUR)
Gap risk	Overnight + weekend	Overnight + weekend	Overnight + weekend	Weekend only
Funding rate cost	5-15% APR	5-12% APR	5-15% APR	3-8% APR
Volatility decay (2x)	2-13%/yr	1-2%/yr	1-4%/yr	0.1-0.7%/yr

7.2 RWA Safety Advantage

RWA assets have significantly lower volatility than crypto, making them inherently safer for leveraged products:

Parameter	RWA Config	Why
Threshold bands	[1.8x, 2.2x]	Standard bands - RWA moves are smaller
Keeper check frequency	Every 5 minutes	Market-hours monitoring sufficient
Ripcord threshold	2.5x	Standard - 12.5% intraday move is extremely rare for RWA
Circuit breaker	3.0x	Standard - 25% single-day move has never occurred for indices
TWAP duration	30 minutes	Standard execution speed
Rebalance cooldown	30 minutes	Standard interval

No major equity index has ever dropped 47.5% in a single day. The worst-ever S&P 500 single-day drop was -20.47% (Black Monday, October 1987). The vault would have survived with 9% margin remaining. For RWA products, the four-layer defense system provides an extraordinary margin of safety that has never been breached in recorded financial history.

7.3 RWA Total Annual Drag

RWA leveraged products have dramatically lower total drag than crypto products due to lower volatility and lower funding costs:

Long 2x Products:

Component	SPX 2x	AAPL 2x	TSLA 2x	GOLD 2x	OIL 2x	EUR 2x
Vol decay	-1.1%	-4.3%	-13.1%	-1.1%	-3.5%	-0.3%
Funding	-8%	-10%	-12%	-8%	-10%	-5%
Rebal costs	-1.5%	-2.0%	-3.0%	-1.5%	-2.0%	-1.0%
Mgmt fee	-1.0%	-1.0%	-1.0%	-1.0%	-1.0%	-1.0%
Total drag	**-11.6%**	**-17.3%**	**-29.1%**	**-11.6%**	**-16.5%**	**-7.3%**

Breakeven analysis (RWA):

- sSPX-2x: S&P 500 must return >5.8%/year (historical average: ~10%)
- sAAPL-2x: Apple must return >8.7%/year
- sGOLD-2x: Gold must return >5.8%/year
- sEUR-2x: EUR/USD must move >3.7%/year

Key insight: RWA leveraged products have total drag of 7-17% for most assets -significantly lower than crypto's 34-64%. Combined with the fact that major indices like the S&P 500 have historically returned ~10%/year, products like sSPX-2x have a realistic path to positive returns after costs. This is the core value proposition.

7.4 Off-Market Hours Handling

RWA assets trade during specific market hours. Winmo handles this with:

Scenario	Handling
During market hours	Full rebalancing active. Keeper monitors every 5 minutes.
After hours (equities)	Last close as anchor. Reduced max leverage. Position held.
Weekends	Friday close as anchor. No rebalancing. Gap risk managed via conservative margin.
Market holidays	Same as weekends. Per-market holiday calendar in governance.
Gap open (Monday)	Immediate rebalance at market open if leverage has drifted.

The 50% margin ratio provides substantial buffer for overnight and weekend gaps. The worst historical overnight gap for the S&P 500 was approximately -5%, which would move leverage from 2.0x to only 2.1x -well within the normal band.

8. Crypto Asset Coverage

8.1 Crypto Asset Characteristics

Crypto assets (BTC, ETH, SOL, etc.) have unique properties that significantly impact leveraged product design:

Property	Value
Trading hours	24/7/365
Annualized volatility	50-120%
Largest single-day move	-37% (BTC) / +42% (BTC)
Gap risk	None (continuous trading)
Execution venue	Hyperliquid (L1)
Funding rate cost	Market-driven (5-50% APR)
Volatility decay (2x)	11-50%+ annual
Price source	Hyperliquid (8-exchange validator median)
Liquidation risk	Higher -requires tighter bands

For comparison, traditional assets (stocks, gold, forex) have lower annualized volatility (8-55%), trade only during market hours, and carry gap risk from overnight/weekend closures. RWA leveraged products (stocks, commodities, indices, forex) also execute on Hyperliquid via its RWA perp pairs (see Section 3.4 for the full product listing).

8.2 Safety Parameters

The protocol uses the following safety parameters for crypto vaults, tuned for crypto's higher volatility:

Parameter	Value	Reason
Threshold bands	[1.85x, 2.15x]	Narrower to catch smaller moves
Keeper check frequency	Every 1 minute	Fast monitoring for 24/7 markets
Ripcord threshold	2.4x	Earlier emergency intervention
Ripcord target	1.3x	More aggressive deleveraging
Circuit breaker	2.7x	Earlier nuclear option
TWAP duration	15 minutes	Faster execution during crash
Max single trade	$2M	Limit slippage on HL
Rebalance cooldown	10 minutes	More frequent rebalances allowed

8.3 Crypto Crash Simulation: BTC -37.5% in 24 Hours (March 2020)

How the vault would have handled the largest single-day BTC drop in history:

```
Starting state: $10M NAV, $20M notional LONG, 2.0x leverage, 50% margin

Hour 0:  BTC at $8,000. Leverage = 2.0x. Margin = 50%.
Hour 1:  BTC drops to $7,760 (-3%). Leverage = 2.06x. OK -within [1.85, 2.15].
Hour 3:  BTC drops to $7,400 (-7.5%). Leverage = 2.16x. □  LAYER 2 TRIGGERED.→
          Rebalance to 2.0x. NAV = $8.5M. New notional = $17M.
Hour 5:  BTC drops to $6,900 (-13.8% from start). Leverage = 2.19x from new base.→
          Cooldown expired. LAYER 2 TRIGGERED again.→
          Rebalance to 2.0x. NAV = $7.3M. New notional = $14.6M.
Hour 8:  BTC drops to $6,200 (-22.5% from start). Leverage = 2.22x.→
          LAYER 2 TRIGGERED. Rebalance. NAV = $6.1M. Notional = $12.2M.
Hour 12: BTC drops to $5,500 (-31.3% from start). Leverage = 2.30x from new base.→
          LAYER 3 RIPCORD eligible (>2.4x). Permissionless caller triggers.→
          Deleverage to 1.3x. NAV = $5.0M. Notional = $6.5M.
Hour 18: BTC drops to $5,000 (-37.5% from start). Leverage = 1.45x from 1.3x base.→
          Still safe. Well within bands.

FINAL: NAV = $4.7M (from $10M = -53% loss).
       Without rebalancing at 2x: NAV = $2.5M (-75% loss).
       Liquidation? NO -margin never dropped below 30%.
```

The vault lost 53% vs 75% without rebalancing, and was never close to liquidation. The multiple rebalance cycles progressively reduced exposure during the crash.

8.4 Crypto Inverse Products: Extra Considerations

Inverse crypto products (sBTC-1xS, sETH-1xS) face the challenge of crypto's extreme upside rallies:

- BTC has risen 42% in a single day (Feb 2021)
- ETH has risen 30% in a single day (Jul 2021)
- SOL has risen 35% in a single day

For inverse crypto vaults, the same tighter parameters apply. Additionally:

Protection	Inverse Crypto Config
Threshold bands	[1.85x, 2.1x] (asymmetrically tighter on upside)
Ripcord threshold	2.3x (earlier trigger)
Circuit breaker	2.5x (very aggressive)
Max position size per vault	$5M (smaller vaults)

Risk warning: Inverse crypto products are the **highest-risk** products in the Winmo lineup. They should be used only for short-term hedging or directional bets, not for extended holding.

8.5 Crypto Execution on Hyperliquid

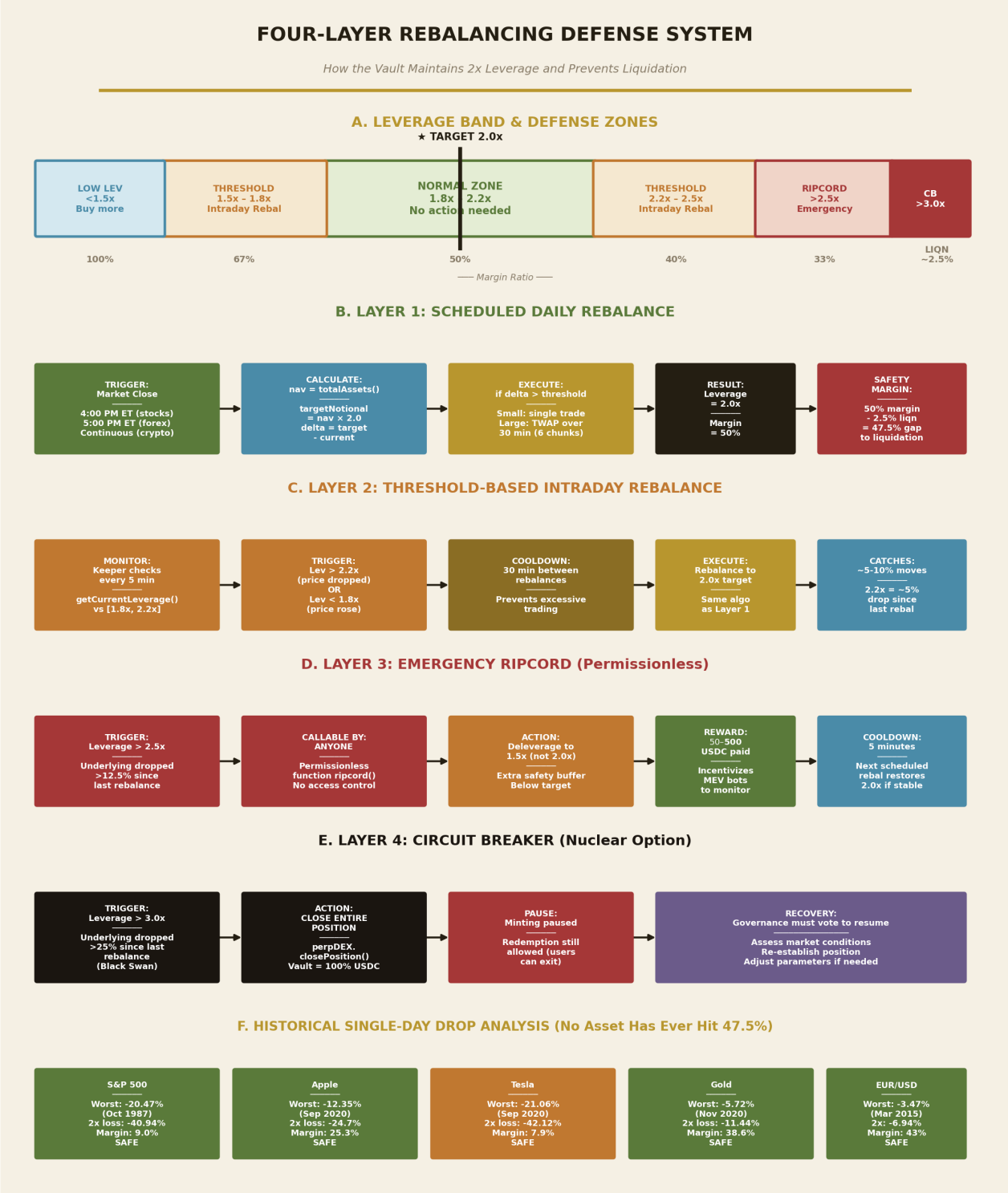
All crypto products execute on Hyperliquid, which provides:

- **24/7 trading:** No market closures, no gap risk.
- **Deep liquidity:** BTC and ETH are the most liquid perp markets in DeFi.
- **Sub-second price updates:** Validator-aggregated median from 8 major exchanges.
- **Built-in MEV protection:** Multi-second on-chain delay for CoreWriter orders.

Available Hyperliquid perp pairs for Winmo crypto products:

Asset	HL Pair	Max Leverage	HL Funding	Est. Annual Drag (2x)
Bitcoin	BTC-USD	50x	5-30% APR	-34% (high vol periods) to -20% (low vol)
Ethereum	ETH-USD	50x	5-25% APR	-42% to -18%
Solana	SOL-USD	20x	10-40% APR	-54% to -24%
Arbitrum	ARB-USD	20x	5-20% APR	-46% to -20%
Chainlink	LINK-USD	20x	5-25% APR	-43% to -20%
Avalanche	AVAX-USD	20x	8-30% APR	-50% to -22%
Dogecoin	DOGE-USD	20x	10-50% APR	-64% to -28%
Sui	SUI-USD	10x	10-40% APR	-55% to -24%

Note: Crypto 2x products have dramatically higher annual drag than traditional leveraged ETFs (e.g., ProShares TQQQ on SPX) due to both higher volatility (more decay) and higher funding rates. Users must understand this.



9. Rebalancing Engine

The rebalancing engine is the core safety system. It operates across four layers of defense:

9.1 Layer 1: Scheduled Daily Rebalance

Trigger: Every day at a fixed time (e.g., 00:00 UTC). Crypto products trade 24/7 with no market close dependency. RWA products (equities, indices, commodities, forex) rebalance at the close of their respective trading sessions.

Algorithm:

```
function scheduledRebalance():
    nav = vault.getNetAssetValue()
    currentNotional = perpDEX.getPositionNotional(vault)
    currentLeverage = currentNotional / nav

    targetNotional = nav × TARGET_LEVERAGE  // 2.0x
    delta = targetNotional - currentNotional

    if abs(delta) < MIN_REBALANCE_THRESHOLD:
        return  // Skip if change is trivial (< 0.5% of notional)

    if abs(delta) > MAX_SINGLE_TRADE:
        // TWAP execution: split into chunks over 30 minutes
        executeTWAP(delta, TWAP_DURATION=30min, TWAP_CHUNKS=6)
    else:
        // Single trade execution
        executeTrade(delta)

    emit Rebalanced(currentLeverage, TARGET_LEVERAGE, delta)
```

Post-rebalance state: Leverage = exactly 2.0x, margin ratio = exactly 50%.

9.2 Layer 2: Threshold-Based Intraday Rebalance

Trigger: When current leverage deviates beyond [1.8x, 2.2x] band during the day.

This catches large intraday moves that would otherwise wait until the scheduled rebalance.

```
function checkAndRebalance():  // Called by keeper every 5 minutes
    currentLeverage = getCurrentLeverage()

    if currentLeverage > UPPER_BAND (2.2x):
        // Underlying dropped → leverage increased → SELL to reduce
        rebalanceToTarget(TARGET_LEVERAGE)

    elif currentLeverage < LOWER_BAND (1.8x):
        // Underlying rose → leverage decreased → BUY to increase
        rebalanceToTarget(TARGET_LEVERAGE)
```

Cooldown: 30-minute minimum between intraday rebalances to prevent excessive trading.

9.3 Layer 3: Emergency Rebalance (Ripcord)

Trigger: Leverage exceeds 2.5x (underlying dropped >12.5% since last rebalance).

Callable by: Anyone (permissionless). Caller receives a USDC reward ($50-$500 depending on vault size) as incentive.

```
function ripcord():  // Callable by ANYONE
    currentLeverage = getCurrentLeverage()
    require(currentLeverage > RIPCORD_THRESHOLD, "Not at ripcord")

    // Emergency deleverage to 1.5x (below target, for safety)
    rebalanceToTarget(RIPCORD_TARGET_LEVERAGE)  // 1.5x

    // Pay caller reward
    usdc.transfer(msg.sender, RIPCORD_REWARD)

    emit Ripcord(msg.sender, currentLeverage, RIPCORD_TARGET_LEVERAGE)
```

Why 1.5x and not 2.0x? During a crash, deleveraging to 1.5x provides extra margin buffer. If the crash continues, the vault has more room before the next ripcord is needed. The next scheduled rebalance restores 2.0x if conditions stabilize.

9.4 Layer 4: Circuit Breaker (Nuclear Option)

Trigger: Leverage exceeds 3.0x (underlying dropped >25% since last rebalance -extreme black swan).

```
function circuitBreaker():  // Callable by keeper or anyone
    currentLeverage = getCurrentLeverage()
    require(currentLeverage > CIRCUIT_BREAKER_THRESHOLD, "Not at CB")

    // Close ENTIRE position → vault holds only USDC
    perpDEX.closePosition(vault)

    // Pause minting (redemption still allowed)
    paused = true

    emit CircuitBreaker(currentLeverage)
    // Governance must vote to resume and re-establish position
```

9.5 Defense-in-Depth Summary

```
LEVERAGE →  1.0x    1.5x    1.8x    2.0x    2.2x    2.5x    3.0x ∞|||||||||||

                     LOWERTARGET |UPPER ||||||
                     BAND |★|       BAND |||||||||||||||

                            OK |||||||←──┼──────┼──→||||||

                    Layer 2: Intraday ||||||
                       Rebalance |||||||||||

                                 Layer 3: Ripcord |||||←──────┼──────┤|||||

                                 Deleverage to 1.5x |||||||||

                                     Layer 4: Circuit |Break |||←──────────┼──────┤||||

                                     Close ALL positions ||||||

      MARGIN: 100%   67%     56%     50%     45%     40%     33%   LIQN
                                                                   (~2.5%)|←──

                                     47.5% GAP to liquidation ──→|
                                    (after daily rebalance)
```

9.6 ADL (Auto-Deleveraging) Handling

If Hyperliquid's insurance fund is depleted during extreme market events, HL triggers Auto-Deleveraging (ADL) -forcibly closing profitable positions to cover bankrupt counterparties. The vault must handle this gracefully:

Detection: The Winmo keeper monitors the vault's position size on Hyperliquid after every block. If the position notional decreases without the vault initiating a trade, an ADL event is detected.

Response mechanism:

1. **Detect ADL:** Keeper detects position reduction not initiated by the vault.
2. **Pause minting:** New deposits are paused to protect incoming users.
3. **Recalculate NAV:** NAV is recalculated based on remaining position + USDC freed by the ADL closure.
4. **Re-establish position:** The vault immediately reopens the closed portion to restore target leverage. The freed USDC is used as collateral for the new position.
5. **Resume normal operations:** Once leverage is restored to target, minting is resumed.

NAV impact: ADL closes the position at Hyperliquid's mark price with no slippage penalty. The

vault's NAV is preserved -only the position structure changes momentarily. Users holding tokens experience no loss beyond normal market movement.

Mitigation: To minimize ADL probability, the vault maintains positions within ADL-safe size ranges (not the largest profitable position on the market). Per-vault position caps at 10% of Hyperliquid OI for the given pair.

9.7 Continuous Market Coverage

Crypto products execute on Hyperliquid, which trades 24/7/365. This eliminates gap risk entirely for crypto vaults -there are no market closures, no overnight gaps, and no weekend gaps. The rebalancing engine operates continuously with no downtime windows for crypto products.

RWA products (equities, indices, commodities, forex) also execute on Hyperliquid via its RWA perp pairs. For these products, gap risk mitigation strategies are in place for assets with trading hours (e.g., equities close at 4 PM ET, forex closes on weekends). While Hyperliquid perp markets may trade 24/7, underlying RWA price discovery is limited to market hours. The rebalancing engine uses market-close-aware scheduling for RWA vaults.

10. Execution Layer: Hyperliquid

10.1 Hyperliquid Overview

Chain: Hyperliquid L1 (custom chain, chain ID 999). Also accessible via Arbitrum bridge.

Collateral: USDC (bridged from Arbitrum).

All Winmo leveraged products execute on Hyperliquid, which provides deep perpetual futures liquidity for both crypto and RWA assets (stocks, commodities, indices, forex) with 24/7 trading, sub-second price updates, and built-in MEV protection.

10.2 Integration Approaches

Two integration approaches are available:

Approach A: HyperEVM Smart Contract (On-Chain)

Deploy the vault directly on HyperEVM. Interact with HyperCore (the perp engine) via system contracts:

```
// Read positions via Precompile at 0x0000...0800
IReadPrecompile(0x0800).perpClearinghouseState(vaultAddress)→
     returns: marginSummary, assetPositions[], crossMaintenanceMarginUsed

// Execute trades via CoreWriter at 0x3333...3333
ICoreWriter(0x3333).sendRawAction(
    abi.encode(
        uint8(1),           // version
        uint24(1),          // action: limit order
        uint32(assetId),    // e.g., BTC-USD
        bool(true),         // isBuy
        uint256(limitPx),   // price
        uint256(sz),        // size
        bool(false),        // reduceOnly
        uint8(tif),         // time in force
        bytes16(cloid)      // client order ID
    )
)
```

Important caveat: CoreWriter order actions have a **multi-second on-chain delay** to prevent MEV/frontrunning. This means rebalancing on Hyperliquid is not atomic.

Approach B: Off-Chain Keeper with API Wallet (Hybrid)

```
Winmo Vault (Arbitrum)│├──

    Holds user USDC + manages ERC-20 tokens on Arbitrum│├──

    Authorized Keeper Bot (off-chain)│├──
            Bridges USDC to Hyperliquid via official bridge│├──
            Places orders via REST API:│
                POST https://api.hyperliquid.xyz/exchange│
                { action: { type: "order", orders: [...] } }│├──
            Monitors positions via:│
                POST https://api.hyperliquid.xyz/info│
                { type: "clearinghouseState", user: "0x..." }│└─
            Reports back to vault contract with signed attestation│└─

    Vault verifies keeper actions via Hyperliquid position state
```

10.3 Fee Impact

Tier	Volume	Taker Fee	Maker Fee
Base	< $5M	0.045%	0.015%
Tier 1	> $5M	0.040%	0.012%
Tier 2	> $25M	0.035%	0.008%
Tier 3	> $100M	0.030%	0.004%

Funding rate cost: Hourly payments. Formula: `F = avg(Premium) + clamp(0.01% − Premium, −0.05%, 0.05%)`. In bullish markets, long funding can be significant (5-30% APR). This is an ongoing cost to the 2x Long ETF product.

10.4 Execution Routing

All products -both crypto and RWA (stocks, commodities, indices, forex) -route to Hyperliquid as the sole execution layer. Hyperliquid supports RWA perp pairs alongside its crypto perp markets, enabling unified execution across all asset classes.

11. Smart Contract Architecture

11.1 Contract Overview

```
┌──────────────────────────────────────────────────────────────────────────────│

                    WINMO LEVERAGED ETF CONTRACTS ||
                         (Deployed on Arbitrum)
                              ||||┌──────────────────────┐┌──────────────────────────────────┐|||


      VaultFactory.sol ||       RebalanceEngine.sol
           ||||──────────────────────||──────────────────────────────||||

      Deploy new vaults ||      Scheduled rebalance logic ||||
      per market pair ||        Emergency rebalance (ripcord) ||||
      Standardized params ||    Circuit breaker |||└──────────┬──────────┘|
        TWAP execution engine
           ||||└──────────────────┬──────────────────┘||▼▼||┌──────────────────────────────────────────────────────────────────────────────
```

```
                LeveragedVault.sol (one per product)
                     ||||────────────────────────────────────────────────────||||

    ERC-4626 compatible deposit/withdraw ||||
    NAV calculation (collateral + unrealized PnL) ||||
    Mint/redeem ERC-20 vault tokens ||||
    Position management on underlying DEX ||||
    Access control for keeper roles
         |||└──────────────────────────────────────────────────────────┘|||||▼||┌──────────────────────────────────────────────────────────┐|||




               HyperliquidAdapter.sol ||||──────────────────────────||||

               Interface to Hyperliquid ||||
               via CoreWriter or API ||||
               Bridge USDC to HL L1 ||||
               Keeper attestations
                    |||└──────────────────────────┘||||┌──────────────────────────────────────────────────────────┐|||



                PriceReader.sol ||||
    Reads position and price data from Hyperliquid ||||
    via ReadPrecompile (0x0800) ||||
    All PnL and leverage calculated from HL prices ||||
    Liquidations determined by HL mark price
         |||└──────────────────────────────────────────────────────────┘|└──────────────────────────────────────────────────────────┘
```

11.2 LeveragedVault.sol -Core Contract

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

import "@openzeppelin/contracts/token/ERC20/extensions/ERC4626.sol";
import "@openzeppelin/contracts/security/ReentrancyGuard.sol";
import "@openzeppelin/contracts/security/Pausable.sol";

contract LeveragedVault is ERC4626, ReentrancyGuard, Pausable {

    // ═══════════════════════════════════════════════════════════
    // STATE
    // ═══════════════════════════════════════════════════════════

    IERC20 public immutable usdc;
    IDEXAdapter public dexAdapter;         // Hyperliquid
    IHyperliquidReader public priceReader;
    IRebalanceEngine public rebalancer;

    uint256 public constant TARGET_LEVERAGE = 2e18;     // 2.0x (18 decimals)
    uint256 public constant UPPER_BAND = 22e17;         // 2.2x
    uint256 public constant LOWER_BAND = 18e17;         // 1.8x
    uint256 public constant RIPCORD_THRESHOLD = 25e17;   // 2.5x
    uint256 public constant RIPCORD_TARGET = 15e17;      // 1.5x
    uint256 public constant CIRCUIT_BREAKER = 3e18;      // 3.0x

    uint256 public constant MIN_REBALANCE_BPS = 50;      // 0.5% minimum delta
    uint256 public constant REBALANCE_COOLDOWN = 30 minutes;
    uint256 public constant RIPCORD_COOLDOWN = 5 minutes;
    uint256 public constant WITHDRAWAL_COOLDOWN = 1 hours;
```

```
    uint256 public ripcordReward;           // USDC reward for ripcord callers
    uint256 public lastRebalanceTimestamp;
    uint256 public lastRipcordTimestamp;

    bytes32 public marketId;                // e.g., keccak256("BTC-USD")
    uint256 public assetIndex;              // Hyperliquid asset index
    bool public isLong;                     // true = 2x Long, false = -1x Inverse

    mapping(address => uint256) public lastDepositTimestamp;

    // ═══════════════════════════════════════════════════════════
    // NAV CALCULATION
    // ═══════════════════════════════════════════════════════════

    /// @notice Total vault assets = USDC balance + unrealized PnL on perp
    function totalAssets() public view override returns (uint256) {
        uint256 usdcBalance = usdc.balanceOf(address(this));
        int256 unrealizedPnl = dexAdapter.getUnrealizedPnL(address(this));
        uint256 collateralOnDEX = dexAdapter.getCollateral(address(this));

        int256 total = int256(usdcBalance + collateralOnDEX) + unrealizedPnl;
        return total > 0 ? uint256(total) : 0;
    }

    /// @notice Current effective leverage
    function getCurrentLeverage() public view returns (uint256) {
        uint256 nav = totalAssets();
        if (nav == 0) return 0;
        uint256 notional = dexAdapter.getPositionNotional(address(this));
        return (notional * 1e18) / nav;
    }

    /// @notice Current margin ratio (collateral + PnL) / notional
    function getMarginRatio() public view returns (uint256) {
        uint256 notional = dexAdapter.getPositionNotional(address(this));
        if (notional == 0) return type(uint256).max;
        uint256 nav = totalAssets();
        return (nav * 1e18) / notional;  // 50% = 5e17 at 2x leverage
    }

    // ═══════════════════════════════════════════════════════════
    // DEPOSIT / WITHDRAW (ERC-4626)
    // ═══════════════════════════════════════════════════════════

    function deposit(uint256 assets, address receiver)
        public override nonReentrant whenNotPaused returns (uint256 shares)
    {
        shares = super.deposit(assets, receiver);
        lastDepositTimestamp[receiver] = block.timestamp;

        // Adjust perp position to maintain target leverage
        _adjustPositionToTarget();
    }

    function withdraw(uint256 assets, address receiver, address owner)
        public override nonReentrant returns (uint256 shares)
    {
        // Enforce withdrawal cooldown (prevents sandwich attacks on rebalance)
        require(
            block.timestamp >= lastDepositTimestamp[owner] + WITHDRAWAL_COOLDOWN,
            "Cooldown active"
        );

        // Reduce perp position first to free collateral
        _reducePositionForWithdrawal(assets);
```

```
        shares = super.withdraw(assets, receiver, owner);
    }

    // ═══════════════════════════════════════════════════════
    // REBALANCING
    // ═══════════════════════════════════════════════════════

    /// @notice Scheduled daily rebalance -keeper only
    function scheduledRebalance() external onlyKeeper {
        _rebalanceToTarget(TARGET_LEVERAGE);
        lastRebalanceTimestamp = block.timestamp;
    }

    /// @notice Threshold-based intraday rebalance -keeper only
    function thresholdRebalance() external onlyKeeper {
        uint256 currentLev = getCurrentLeverage();
        require(
            currentLev > UPPER_BAND || currentLev < LOWER_BAND,
            "Within band"
        );
        require(
            block.timestamp >= lastRebalanceTimestamp + REBALANCE_COOLDOWN,
            "Cooldown"
        );

        _rebalanceToTarget(TARGET_LEVERAGE);
        lastRebalanceTimestamp = block.timestamp;
    }

    /// @notice Emergency ripcord -callable by ANYONE
    function ripcord() external nonReentrant {
        uint256 currentLev = getCurrentLeverage();
        require(currentLev > RIPCORD_THRESHOLD, "Below ripcord");
        require(
            block.timestamp >= lastRipcordTimestamp + RIPCORD_COOLDOWN,
            "Ripcord cooldown"
        );

        _rebalanceToTarget(RIPCORD_TARGET);  // Deleverage to 1.5x
        lastRipcordTimestamp = block.timestamp;

        // Reward caller
        usdc.transfer(msg.sender, ripcordReward);
        emit Ripcord(msg.sender, currentLev, RIPCORD_TARGET);
    }

    /// @notice Circuit breaker -callable by ANYONE
    function circuitBreaker() external nonReentrant {
        uint256 currentLev = getCurrentLeverage();
        require(currentLev > CIRCUIT_BREAKER, "Below CB threshold");

        dexAdapter.closeAllPositions(address(this));
        _pause();  // Pause minting; redemption still allowed

        emit CircuitBreakerTriggered(currentLev);
    }

    // ═══════════════════════════════════════════════════════
    // INTERNAL POSITION MANAGEMENT
    // ═══════════════════════════════════════════════════════

    function _rebalanceToTarget(uint256 targetLev) internal {
        uint256 nav = totalAssets();
        uint256 currentNotional = dexAdapter.getPositionNotional(address(this));
```

```
        uint256 targetNotional = (nav * targetLev) / 1e18;

        if (targetNotional > currentNotional) {
            // Need to INCREASE position (buy more)
            uint256 delta = targetNotional - currentNotional;
            if (_isAboveMinThreshold(delta, currentNotional)) {
                dexAdapter.increasePosition(delta);
            }
        } else {
            // Need to DECREASE position (sell some)
            uint256 delta = currentNotional - targetNotional;
            if (_isAboveMinThreshold(delta, currentNotional)) {
                dexAdapter.decreasePosition(delta);
            }
        }

        emit Rebalanced(currentNotional, targetNotional, targetLev);
    }

    function _adjustPositionToTarget() internal {
        if (totalAssets() > 0) {
            _rebalanceToTarget(TARGET_LEVERAGE);
        }
    }

    function _isAboveMinThreshold(uint256 delta, uint256 notional)
        internal pure returns (bool)
    {
        return (delta * 10000) / notional >= MIN_REBALANCE_BPS;
    }
}
```

11.3 Keeper Infrastructure

Rebalancing is triggered by keepers. **Winmo operates the primary keeper** with Gelato Functions and custom bots as redundant backups:

```
// Gelato task registration
GelatoAutomate.createTask({
    execAddress: vault,
    execSelector: vault.thresholdRebalance.selector,
    moduleData: ModuleData({
        modules: [Module.RESOLVER],
        args: [resolverAddress, checkSelector]
    })
});

// Resolver contract
contract RebalanceResolver {
    function checker() external view returns (bool canExec, bytes memory execPayload) {
        uint256 leverage = vault.getCurrentLeverage();

        if (leverage > UPPER_BAND || leverage < LOWER_BAND) {
            canExec = true;
            execPayload = abi.encodeCall(vault.thresholdRebalance, ());
        }

        // Also check scheduled rebalance timing
        if (block.timestamp >= vault.lastRebalanceTimestamp() + DAILY_INTERVAL) {
            canExec = true;
            execPayload = abi.encodeCall(vault.scheduledRebalance, ());
        }
    }
}
```

12. Mathematical Framework

12.1 Leveraged Return Formula

For a daily-rebalanced leveraged product with leverage factor L, if the underlying has daily return r_t on day t:

Single-day return (Long 2x, L = +2):

```
R_vault(t) = L × r_t − fees(t)
```

Single-day return (Inverse -1x, L = −1):

```
R_vault(t) = −1 × r_t − fees(t)
```

When the underlying rises 5%: Long 2x returns +10%, Inverse -1x returns −5%.

When the underlying falls 5%: Long 2x returns −10%, Inverse -1x returns +5%.

Multi-day cumulative return (N days):

```
V(N)/V(0) = ∏(t=1 to N) [1 + L × r_t − f_t]
```

where f_t = daily fee accrual (funding rate + management fee + rebalancing cost).

This is NOT equal to L × (S(N)/S(0) − 1). The product is path-dependent. This applies to BOTH long and inverse products.

12.2 Volatility Decay (Variance Drain)

Assuming the underlying follows geometric Brownian motion with drift μ and volatility σ:

```
dS/S = μ dt + σ dW
```

The leveraged product's continuous-time value satisfies:

```
V(T) = V(0) × (S(T)/S(0))^L × exp−[(L² − L) ×σ ² × T / 2]
```

The decay term: `exp[−−(L² −− L) × σ² × T / 2]`

For L = +2 (Long 2x): `(L² −− L) = (4 −− 2) = 2` → Decay = $\exp[-\sigma^2 \times T]$

For L = +3 (Long 3x): `(L² −− L) = (9 −− 3) = 6` → Decay = $\exp[-3\sigma^2 \times T]$

For L = −1 (Inverse -1x): `(L² −− L) = (1 −− (−−1)) = 2` → Decay = $\exp[-\sigma^2 \times T]$

For L = −2 (Inverse -2x): `(L² −− L) = (4 −− (−−2)) = 6` → Decay = $\exp[-3\sigma^2 \times T]$

Key insight: Long 2x and Inverse -1x share the same decay rate. Long 3x and Inverse -2x share the same (3x higher) decay rate.

Annual decay estimates across all leverage tiers:

Asset	σ (annual)	Long 2x	Long 3x	Inverse -1x	Inverse -2x
S&P 500	**~15%**	**−1.1%**	**−3.3%**	**−1.1%**	**−3.3%**
Apple	**~30%**	**−4.3%**	**−12.2%**	**−4.3%**	**−12.2%**
Gold	**~15%**	**−1.1%**	**−3.3%**	**−1.1%**	**−3.3%**
EUR/USD	**~8%**	**−0.3%**	**−0.9%**	**−0.3%**	**−0.9%**
Tesla	**~55%**	**−13.1%**	**−34.2%**	**−13.1%**	**−34.2%**
Bitcoin	~60%	−15.1%	−38.3%	−15.1%	−38.3%
Ethereum	~75%	−21.6%	−51.3%	−21.6%	−51.3%
Solana	~90%	−27.9%	−62.1%	−27.9%	−62.1%

RWA assets have dramatically lower decay than crypto. The S&P 500 at ~15% annual volatility loses only 1.1% per year to decay on a 2x product -making sSPX-2x viable for medium-term holding. By contrast, crypto assets with 60-90% volatility lose 15-28% annually to decay alone.

Interpretation: For RWA products like sSPX-2x, total annual drag is approximately 11.6% (see Section 7.3). With the S&P 500's historical average return of ~10%, the product has a realistic path to positive returns. Crypto products require much larger underlying moves to overcome their higher drag.

12.3 Funding Rate Impact

On Hyperliquid, the vault pays **hourly funding** -a market-driven rate that reflects the imbalance between longs and shorts.

Annualized funding cost model:

```
Long 2x:    Cost = Funding Rate × Notional × Time (typically pays in bull markets)
Inverse −1x:  Cost = Funding Rate × Notional × Time (may receive in bull markets)
```

Key difference for inverse products: On Hyperliquid, when the market is bullish (more longs than shorts), **shorts receive funding**. This means inverse products can have **negative** funding costs (a tailwind) during bull markets.

Product	Funding in Bull Market	Funding in Bear Market
Long 2x	Pays 10-30% APR	Pays 5-15% APR
Inverse -1x	**Receives 10-30% APR**	**Pays 5-15% APR**

Example: sBTC-1xS (inverse BTC) during a bull market:

- Notional: $20M SHORT
- Funding: Shorts receive ~15% APR on Hyperliquid
- Annual funding income: $20M × 15% = $3,000,000 = **+30% of NAV**
- This offsets much of the volatility decay

12.4 Total Expected Annual Drag

RWA Long 2x Products (primary focus):

Component	SPX 2x	AAPL 2x	TSLA 2x	GOLD 2x	OIL 2x	EUR 2x
Vol decay	−1.1%	−4.3%	−13.1%	−1.1%	−3.5%	−0.3%
Funding	−8%	−10%	−12%	−8%	−10%	−5%
Rebal costs	−1.5%	−2.0%	−3.0%	−1.5%	−2.0%	−1.0%
Mgmt fee	−1.0%	−1.0%	−1.0%	−1.0%	−1.0%	−1.0%
Total drag	**−11.6%**	**−17.3%**	**−29.1%**	**−11.6%**	**−16.5%**	**−7.3%**

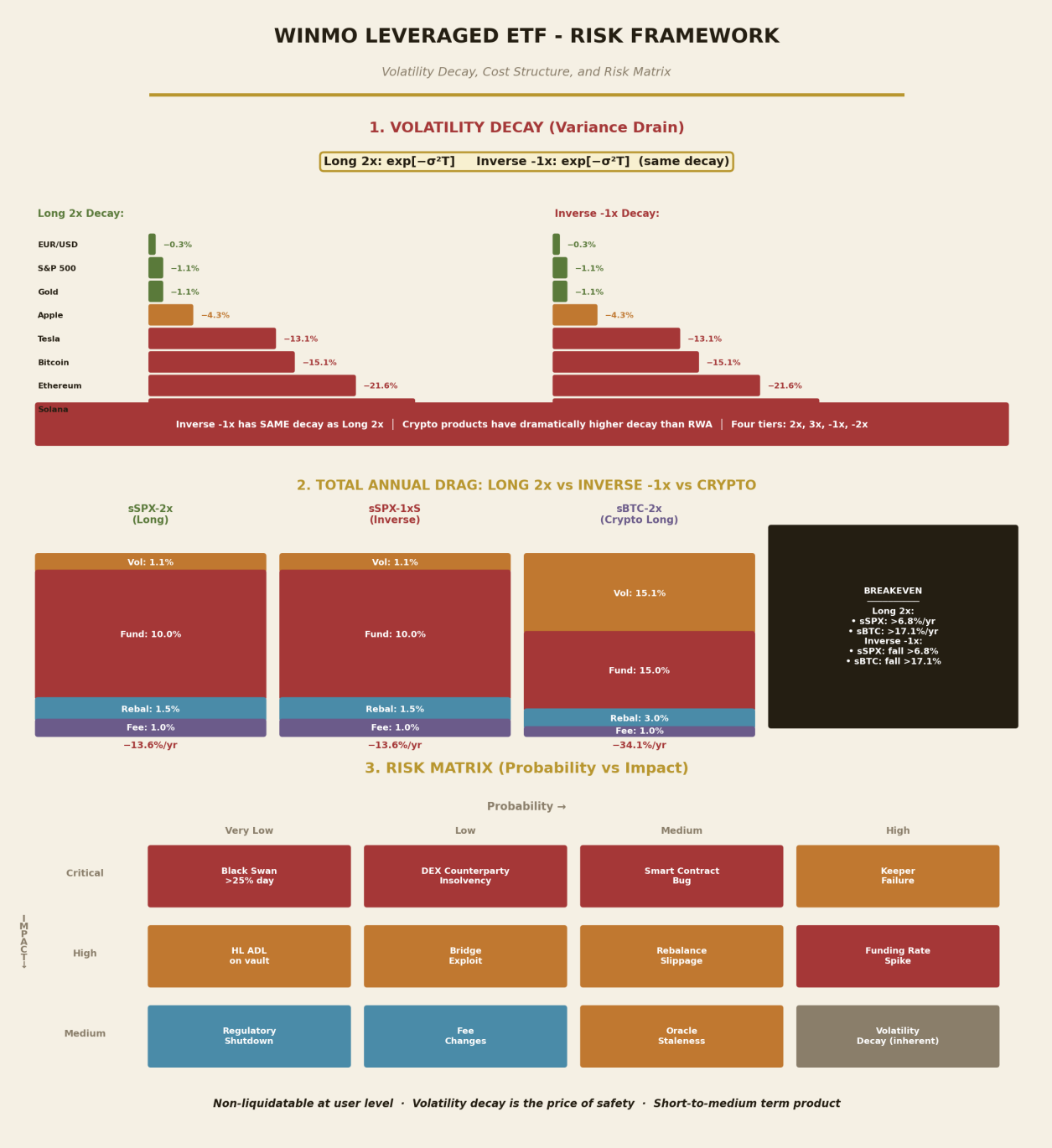
Crypto Long 2x Products:

Componei	BTC 2x	ETH 2x	SOL 2x	ARB 2x	LINK 2x	AVAX 2x	DOGE 2x	SUI 2x
Vol decay	−15.1%	−21.6%	−27.9%	−25.0%	−22.5%	−26.0%	−32.5%	−29.0%
Funding	−15%	−15%	−20%	−15%	−15%	−18%	−25%	−20%
Rebal costs	−3.0%	−4.0%	−5.0%	−4.5%	−4.0%	−4.5%	−5.0%	−5.0%
Mgmt fee	−1.0%	−1.0%	−1.0%	−1.0%	−1.0%	−1.0%	−1.0%	−1.0%
Total drag	**−34.1%**	**−41.6%**	**−53.9%**	**−45.5%**	**−42.5%**	**−49.5%**	**−63.5%**	**−55.0%**

Key takeaway: RWA products have 3-5x lower total drag than crypto products. sSPX-2x at -11.6% annual drag is comparable to ProShares TQQQ in TradFi, while crypto products face 34-64% annual drag. RWA leveraged products are the core value proposition for medium-term directional exposure.

Breakeven analysis:

- sSPX-2x (Long): S&P 500 must return >5.8%/year (historical avg: ~10% - **above breakeven**)
- sAAPL-2x (Long): Apple must return >8.7%/year
- sGOLD-2x (Long): Gold must return >5.8%/year
- sBTC-2x (Long): BTC must return >17.1%/year
- sETH-2x (Long): ETH must return >20.8%/year



13. Protocol Risk Analysis

13.1 Smart Contract Risk

Risk	Severity	Mitigation	
Vault contract bug (reentrancy, accounting error)	Critical	3+ independent audits (Trail of Bits, OpenZeppelin, Spearbit). Formal verification of NAV and rebalance logic. $1M+ Immunefi bug bounty.	
Hyperliquid L1 bug	Critical	HL is a custom L1 with limited battle-testing vs Ethereum. Mitigation: monitor HL status and upgrades; circuit breaker auto-triggers if execution cannot be confirmed.	
ERC-4626 rounding exploit	Medium	Use OpenZeppelin's audited ERC-4626 implementation with virtual offset for inflation attack protection.	

13.2 Price / Execution Risk

Risk	Severity	Mitigation	
Price manipulation on Hyperliquid	Medium	HL aggregates prices from 8 major exchanges via stake-weighted validator median. Manipulation requires corrupting multiple validators simultaneously.	
Hyperliquid price staleness	Medium	HL validators update prices sub-second. If price feeds stall, the vault pauses rebalancing until fresh data is confirmed.	
Mark price vs index price divergence	Low	Liquidations on HL are based on mark price. The vault monitors mark price directly since that determines the vault's actual margin ratio.	

13.3 Execution / Rebalancing Risk

Risk	Severity	Mitigation	
Keeper failure (no rebalance executed)	High	Winmo operates its own keeper as primary. Gelato and custom bots as backup. Ripcord is permissionless (anyone can call). Financial incentive ($50-$500 reward).	
Rebalance slippage (large orders move price)	Medium	TWAP execution for large rebalances (split into chunks over 15 min). Slippage caps (0.5% max per trade).	
TWAP adversarial frontrunning	Medium	Randomized TWAP intervals (not fixed schedule). Hyperliquid has built-in MEV protection (multi-second delay).	
Rebalance during low liquidity	Medium	Vault checks Hyperliquid order book depth before executing. If insufficient depth, reduces trade size or splits.	

13.4 Counterparty / Protocol Risk

Risk	Severity	Mitigation	
Hyperliquid ADL (auto-deleveraging)	High	Profitable vault positions can be forcibly closed if losing traders go bankrupt and insurance is depleted. Mitigation: maintain positions within ADL-safe size range.	
Protocol shutdown / exit scam	Medium	Hyperliquid is a custom L1 (more trust required than Ethereum). Mitigation: monitor HL governance and validator set; maintain emergency withdrawal procedures.	
Fee structure changes	Low	Hyperliquid could increase fees via governance. Mitigation: monitor governance proposals; emergency withdraw if fees spike.	

13.5 Economic / Market Risk

Risk	Severity	Mitigation	
Volatility decay	Inherent	Cannot be eliminated. Clear user disclosure. Product is designed for short-to-medium term holding. Display real-time decay metrics on front-end.	
Funding rate spike	High	In extreme bull markets, long funding can reach 50-100% APR on Hyperliquid. Mitigation: display current annualized funding cost. Governance can pause products where funding exceeds threshold.	
Negative funding windfall	Positive	In bear markets, short/inverse products (-1xS) benefit from negative funding (shorts get paid). This is a tailwind.	
Black swan drop (>25% single day)	Low prob / High impact	Circuit breaker closes position at 2.7x.	
Black swan rally (>25% single day)	Low prob / High impact	Affects inverse products. Circuit breaker closes inverse position.	
Cascading liquidations on underlying DEX	Medium	If many leveraged traders on Hyperliquid are liquidated simultaneously, the vault's position may be affected by ADL or slippage. Mitigation: OI cap awareness; don't be the largest position on the DEX.	

13.5a Inverse-Specific Risk

Risk	Severity	Mitigation	
Volatility decay (same as long 2x)	Inherent	Inverse -1x decay = exp[-sigma^2 T], same as long 2x. Clear disclosure. Short-to-medium term recommended.	
Unlimited upside of underlying	High	Crypto has no price ceiling. Prolonged bull market can destroy inverse NAV to near zero. Mitigation: prominent warnings, frontend decay calculator.	
Short squeeze on underlying DEX	High	If the vault's short is a significant portion of OI, a squeeze can cause extreme slippage during rebalancing. Mitigation: per-vault position caps at 10% of DEX OI.	
Inverse crypto rally risk	Critical	Crypto can rally 30-40%+ in a day. Inverse crypto vaults use tighter circuit breakers (2.5x). Maximum per-vault size capped at $5M.	
Funding rate reversal	Medium	Inverse products may benefit from funding in bull markets but pay in bear markets. If the inverse product exists to hedge a bear, funding becomes a cost precisely when the product should profit.	

13.5b Crypto-Specific Risk

Risk	Severity	Mitigation	
Extreme volatility (60-120% annual sigma)	Inherent	Tighter rebalancing bands [1.85x, 2.15x]. 1-minute keeper checks. Decay clearly disclosed.	
Crypto flash crash (>30% intraday)	High	Layer 2 threshold triggers ~7-10 times during a 30% crash. Each rebalance reduces exposure. Ripcord at 2.4x.	
Hyperliquid L1 downtime	High	HL is a custom L1 with less battle-testing than Ethereum. If HL goes down, crypto vaults cannot rebalance. Mitigation: circuit breaker auto-triggers if keeper cannot confirm execution.	
Crypto funding rate volatility	High	HL funding can swing from +50% to -30% APR within weeks. Drag estimates vary wildly. Mitigation: frontend displays real-time annualized funding.	
Altcoin liquidity risk	High	SOL, ARB, AVAX, DOGE have lower HL perp liquidity than BTC/ETH. Large rebalances may face >1% slippage. Mitigation: smaller max vault sizes ($2-5M) for altcoins. Per-trade slippage caps.	
Altcoin delisting risk	Medium	Hyperliquid may delist low-volume perp pairs. If the underlying perp market is removed, vault must close position. Mitigation: monitor HL governance; graceful wind-down procedure.	

13.6 Bridge / Cross-Chain Risk

Risk	Severity	Mitigation	
Arbitrum bridge exploit	Critical	Vault contracts are on Arbitrum; USDC must be bridged to Hyperliquid for execution. Mitigation: limit USDC bridged; maintain reserves on Arbitrum.	
Hyperliquid bridge exploit	High	HL bridge depends on Arbitrum bridge security. ~5 min finalization. Mitigation: limit USDC bridged to HL; maintain sufficient reserves on Arbitrum for redemptions.	
Bridge delay (7-day Arbitrum L2→L1)	Medium	For cross-chain redemptions. Mitigation: maintain USDC reserves on Arbitrum for redemptions.	

13.7 Governance / Operational Risk

Risk	Severity	Mitigation
Keeper key compromise	High	Keeper can only call rebalance functions (bounded actions). Cannot withdraw user funds. Multi-sig for critical admin functions.
Governance attack (malicious parameter change)	Medium	Timelock on all parameter changes (48h minimum). Community monitoring.
Regulatory shutdown	Medium	Protocol is permissionless at contract level. Frontend can be geo-blocked. Users can always interact directly with contracts.

13.8 Risk Matrix Summary

```
PROBABILITY →     Very Low        Low          Medium       High||||

IMPACT ↓||||||||

Critical         Black Swan     HL L1         Smart       Keeper
                 (>25% day)     Downtime      Contract    Failure
                                              Bug||||

High             HL ADL          Bridge       Rebalance    Funding
                 on vault       Exploit       Slippage     Rate
                                                           Spike||||

Medium           Regulatory     Fee           Price         Vol
                 Shutdown       Changes       Staleness     Decay
                                                          (inherent)
```

14. Fee Structure & Economics

14.1 User-Facing Fees

Fee	Rate	Recipient
Mint fee	0.1% of deposit	Protocol treasury
Redeem fee	0.1% of withdrawal	Protocol treasury
Management fee	1.0% per year (accrued daily to NAV)	50% stakers, 50% treasury

14.2 Internal Costs (Absorbed by NAV)

Cost	Estimated Annual Impact	Source
Hyperliquid funding	5-30%	Market-driven hourly funding rate
Hyperliquid trading fees	0.3-0.8%	0.03-0.045% × rebalance trades
Rebalancing slippage	0.3-1.0%	Market impact during rebalance
Keeper gas	<0.1%	Arbitrum L2 gas (cheap)

14.3 Revenue Model at Scale

AUM	Management Fee	Mint/Redeem Fees (10% turnover/mo)	Total Revenue
$10M	$100K	$240K	$340K
$50M	$500K	$1.2M	$1.7M
$100M	$1M	$2.4M	$3.4M
$500M	$5M	$12M	$17M

15. Governance & Operations

15.1 Governance-Controlled Parameters

Parameter	Default	Range	Timelock
Target leverage	2.0x	1.5x – 3.0x	48 hours
Rebalance bands	[1.8x, 2.2x]	[1.5x, 2.5x]	48 hours
Ripcord threshold	2.5x	2.3x – 2.8x	48 hours
Circuit breaker	3.0x	2.8x – 3.5x	48 hours
Ripcord reward	$100	$50 – $500	24 hours
Management fee	1.0%	0 – 2.0%	7 days
Withdrawal cooldown	1 hour	0 – 24 hours	48 hours
New product listing	N/A	Governance vote	48 hours

15.2 Operational Roles

Role	Permissions	Trust Level
Keeper (Winmo primary + Gelato backup)	Call rebalance functions only. Cannot withdraw funds.	Low trust (bounded actions)
Guardian (3-of-5 multisig)	Emergency pause. Cannot modify parameters.	Medium trust
Governance (WINMO token holders)	All parameter changes via timelock. Product listings.	Decentralized
Anyone	Call ripcord(), circuit-Breaker()	Zero trust (permissionless)

16. Roadmap

Phase	Timeline	Deliverables	
Phase 1: Design, Audit & Testnet	Q2 2026	Smart contract development. 3 independent audits. Formal verification of rebalance logic. Testnet deployment on Arbitrum Sepolia. Paper trading with historical data backtests.	
Phase 2: Mainnet Beta + Inverse	Q3 2026	Long 2x products: sSPX-2x, sGOLD-2x, sBTC-2x, sETH-2x. Inverse -1x products: sSPX-1xS, sGOLD-1xS, sBTC-1xS, sETH-1xS. $5M AUM cap per vault. Execution on Hyperliquid.	
Phase 3: Full Suite & Scale	Q4 2026	All 2x long + -1x inverse products live. Long expansion: sAAPL-2x, sTSLA-2x, sNVDA-2x, sOIL-2x, sSOL-2x. 3x long and -2x inverse tiers launch. Altcoin vaults. Remove AUM caps. $100M+ AUM target. Custom leverage vaults. DAO governance.	

17. Conclusion

The Winmo Leveraged ETF Protocol creates a product category that does not yet exist in DeFi: **permissionless, non-liquidatable, daily-rebalanced multi-leverage (2x, 3x, -1x, -2x) exposure to real-world asset prices and cryptocurrency.**

The protocol offers four leverage tiers:

Family	Leverage	Example	Description	
Long 2x	+2x	sAAPL-2x, sBTC-2x	Standard leveraged long exposure	
Long 3x	+3x	sAAPL-3x, sBTC-3x	Aggressive leveraged long exposure	
Inverse -1x	-1x	sAAPL-1xS, sBTC-1xS	Standard inverse (short) exposure	
Inverse -2x	-2x	sAAPL-2xS, sBTC-2xS	Leveraged inverse (short) exposure	
Asset classes	5	Equities, Indices, Commodities, Forex, Crypto	Full cross-asset coverage	

The mechanism achieves non-liquidatability through a four-layer defense system (scheduled rebalance → threshold rebalance → ripcord → circuit breaker) that ensures the underlying perp position's margin ratio never approaches the liquidation zone on Hyperliquid. All vaults use asset-class-tuned safety parameters optimized for each asset's volatility profile.

Inverse products use identical vault architecture with mirrored position direction. The same rebalancing and safety system applies -after losses (underlying moves against the position), the vault reduces exposure. Higher leverage tiers (3x, -2x) experience proportionally more volatility decay, while -1x inverse has the same decay rate as long 2x.

The trade-off for all product families is volatility decay -a mathematically unavoidable property of all leveraged products that rebalance. Decay scales with leverage: higher tiers (3x, -2x) suit shorter holding periods, while lower tiers (2x, -1x) are viable for medium-term directional bets. These products are designed for short-to-medium term exposure, not buy-and-hold.

By building on Hyperliquid as the sole execution layer for both crypto and RWA assets, Winmo inherits its price infrastructure, deep liquidity, and market coverage while adding the vault, rebalancing, and tokenization layer that transforms a raw perp position into a simple, non-liquidatable ERC-20 token. Hyperliquid's support for RWA perp pairs (stocks, commodities, indices, forex) alongside crypto enables unified execution across all asset classes under a single protocol.

18. References

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19. Disclaimer

This whitepaper is for informational purposes only and does not constitute an offer to sell, a solicitation of an offer to buy, or a recommendation of any token, investment product, or service.

Leveraged products amplify both gains and losses. A 2x leveraged product will lose approximately 2x the daily percentage decline of the underlying asset. Over time, volatility decay will erode value even if the underlying is flat. Users can lose a significant portion or all of their investment.

Higher leverage tiers (3x, -2x) carry significantly more risk. 3x long and -2x inverse products experience 3x the volatility decay of 2x/-1x products. **Inverse products carry directional risk** -the underlying asset has theoretically unlimited upside, meaning inverse products can lose significant value in sustained bull markets. Higher leverage products should be held for short-term bets only.

Cryptocurrency assets have high volatility. Winmo products cover both RWA and crypto assets. Crypto products carry significantly higher volatility (60-120% annual sigma). Single-day moves exceeding 30% have occurred historically for BTC, ETH, and SOL. The protocol relies heav-

ily on intraday rebalancing layers and uses tighter safety parameters tuned for crypto's volatility profile.

The "non-liquidatable" property applies to the end user only. The underlying vault positions on Hyperliquid can theoretically be liquidated in extreme scenarios (e.g., >47.5% intraday move combined with complete keeper failure and ripcord failure). With crypto-tuned tighter bands, the tolerance is lower. The four-layer defense system is designed to prevent this, but no system is infallible.

Smart contract risk is real. Despite audits and formal verification, smart contract bugs can result in total loss of funds. The protocol interacts with external contracts (Hyperliquid, bridges) which expand the attack surface.

Price risk is real. The protocol depends on accurate prices from Hyperliquid's validator median (aggregated from 8 major exchanges). Hyperliquid price manipulation or stale feeds could result in incorrect rebalancing and loss of funds.

Past performance is not indicative of future results. Backtested returns do not account for all real-world costs, slippage, and failure modes.

Users are solely responsible for understanding these risks and ensuring compliance with applicable laws in their jurisdiction. Nothing in this whitepaper constitutes financial, legal, or investment advice.

Spot. Leveraged. Inverse. Every Asset Onchain. The leveraged layer for all of DeFi.

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